

Week 3

Category / Ratio	2023	2022	2021
PROFITABILITY			
Vertical Analysis (% of Net Sales)			
Sales	100.0%	100.0%	100.0%
Cost of Goods Sold	69.0%	70.0%	73.5%
Gross Income (Profit)	31.0%	30.0%	26.5%
Bad Debt Expense	0.3%	0.1%	1.4%
General & Admin Expense	3.1%	3.5%	3.8%
Salaries	13.5%	13.9%	14.3%
Selling Expense	4.0%	3.8%	3.9%
Depreciation	1.3%	1.1%	1.4%
Income from Operations	↑ 8.7%	7.5%	1.9%
Other Income	1.4%	2.3%	2.0%
Interest Expense	1.3%	0.9%	1.3%
Income Before Tax	8.8%	8.9%	2.6%
Income After Tax (Net Profit Margin)	↑ 4.7%	4.8%	1.3%
Return on Average Equity (ROE)	18.9%	16.5%	N/A 4.2%
EFFICIENCY			
Age of Receivables	72.1 days	70.7 days	71.7 days
Age of Inventory (using COGS)	↑ 117.6 days	111.7 days	120.6 days
Age of Payables (using Purchases)*	↑ 52.3 days	↑ 30.8 days	23.3 days
LIQUIDITY			
Current Ratio	1.75 : 1	2.06 : 1	2.02 : 1

Not A Big focus

most recent

Now we need more loans

1. Profitability Ratios

These measure a company's ability to generate income relative to revenue, assets, or equity.

- Gross Profit Margin:** Measures percentage of revenue remaining after deducting direct production costs.

$$\text{Gross Profit Margin} = \frac{\text{Gross Profit}}{\text{Net Sales}}$$

- Net Profit Margin:** Percentage of revenue left after paying all operating expenses, interest, and taxes.

$$\text{Net Profit Margin} = \frac{\text{Net Income}}{\text{Net Sales}}$$

- Return on Equity (ROE):** Indicates how effectively the company uses shareholders' capital to generate profit.

$$\text{ROE} = \frac{\text{Net Income}}{\text{Shareholders' Equity}}$$

2. Liquidity Ratios

These assess a firm's ability to cover its short-term debt obligations without raising outside capital.

- Current Ratio:** Measures whether short-term assets can cover short-term liabilities.

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

- Quick (Acid-Test) Ratio:** A stricter liquidity test that excludes inventory (less liquid asset).

$$\text{Quick Ratio} = \frac{\text{Cash} + \text{Accounts Receivable} + \text{Short-Term Investments}}{\text{Current Liabilities}}$$

3. Efficiency & Activity Ratios

These evaluate how productively a company manages its resources and working capital.

- Age of Inventory / Inventory Turnover:** Measures how quickly inventory is converted into sales.

$$\text{Age of Inventory (Days)} = \left(\frac{\text{Inventory}}{\text{COGS}} \right) \times 365$$

Age of Receivables (Collection Period): Measures average number of days taken to collect customer receivables.

$$\text{Age of Receivables (Days)} = \left(\frac{\text{Accounts Receivable}}{\text{Net Sales}} \right) \times 365$$

- Age of Payables:** Measures average number of days taken to pay trade suppliers.

$$\text{Age of Payables (Days)} = \left(\frac{\text{Accounts Payable}}{\text{Purchases}} \right) \times 365$$

4. Solvency & Leverage Ratios

These measure long-term financial stability and a firm's reliance on debt to finance operations.

- Debt-to-Equity (D/E):** Compares total debt liabilities against total owner equity.

$$\text{Debt-to-Equity} = \frac{\text{Total Liabilities}}{\text{Total Shareholders' Equity}}$$

- Interest Coverage Ratio:** Evaluates how easily earnings cover annual interest payments.

$$\text{Interest Coverage} = \frac{\text{EBIT}}{\text{Interest Expense}}$$

5. Growth Ratios

These track period-over-period percentage changes to evaluate company expansion rates.

- Sales Growth:** Percentage change in top-line revenue over a specified period.

$$\text{Sales Growth} = \frac{\text{Sales}_{\text{Current}} - \text{Sales}_{\text{Prior}}}{\text{Sales}_{\text{Prior}}} \times 100$$

- Net Income Growth:** Percentage change in net bottom-line earnings over a specified period.

38.6%

Current ratio

= current Assets $\div$ current liabilities

eg. 2:1

Acid test

= Cash + M/S + A/R $\div$ current liability

eg. 1:1

debt to equity

eg. 3:1

Net worth to total Assets

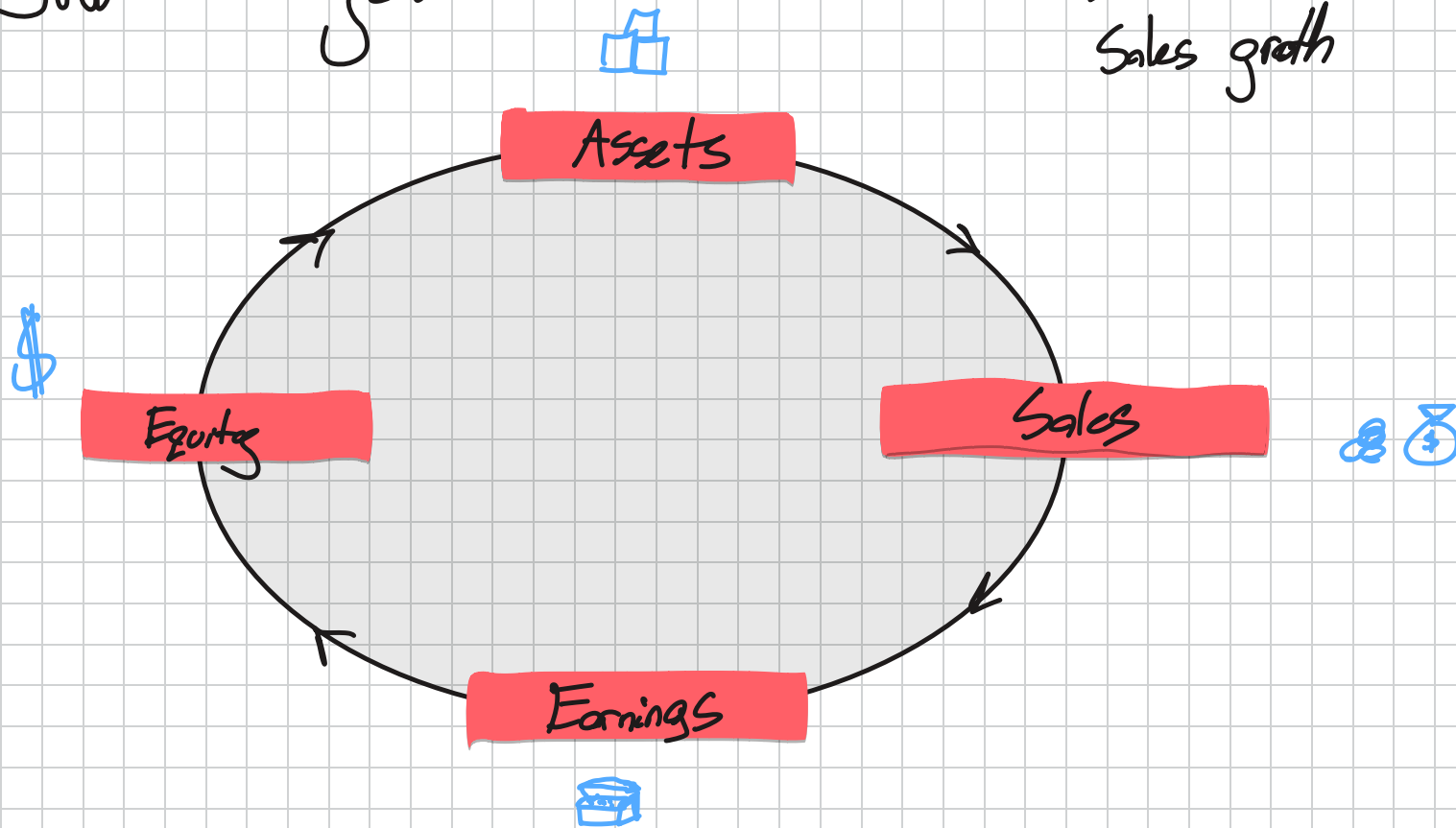
Eg. 25%

interest coverage

Eg. high as possible

Growth Cycle

38.6%
Sales growth



Scenario 1: Owner's Projections (\$750,000 Sales)

Part 1: Projected Income Statement (P&L)
We start with the sales figure provided by the owner.

- "Sales": Given as "\$750,000".
- "Cost of Goods Sold (COGS)": Gross profit is 20% of sales, meaning COGS is 80%.
Calculation: $\$750,000 \times 0.80 = \$600,000$.
- "Gross Profit": $\$750,000 - \$600,000 = \$150,000$.
- "Operating Expenses":
 - "Salary Expense": Given as a flat "\$42,000".
 - "Depreciation Expense": Given as "\$6,000".
 - "Other Operating Expenses": 8% of sales.
Calculation: $\$750,000 \times 0.08 = \$60,000$.
 - "Total Operating Expenses": $\$42,000 + \$6,000 + \$60,000 = \$108,000$.
- "Operating Income (Net Before Tax)": Gross Profit - Total OpEx.
Calculation: $\$150,000 - \$108,000 = \$42,000$.
- "Income Tax Expense": Item 6 says tax is 25% of Net Income Before Tax.
Calculation: $\$42,000 \times 0.25 = \$10,500$.
- "Net Income After Tax": $\$42,000 - \$10,500 = \$31,500$.
- "Dividends": Item 17 says dividends are 25% of Net Income after tax.
Calculation: $\$31,500 \times 0.25 = \$7,875$.

Review recording
- you have errors

Part 2: Projected Balance Sheet (End of Year)

"Assets":

- "Accounts Receivable": Item 8 says "10 days' sales". Using our 360-day standard ($10/360$):
Calculation: $\$750,000 / 360 \times 10 = \$20,833$.
- "Inventory": Item 4 implies a 90-day inventory age based on COGS ($90/360 = 1/4$ year).
Calculation: $\$600,000 \text{ (COGS)} / 4 = \$150,000$. (Note: This matches last year's ending inventory because sales didn't change volume.)
- "Land": Item 10 cost is "\$30,000".
- "Buildings and Fixtures (Net)": Cost minus Accumulated Depreciation.
Accumulated Depreciation = End of Year Balance (\$21,000 from Item 12) + This Year's Expense (\$6,000 from Item 13) = \$27,000.
Net Book Value: $\$80,000 \text{ (Cost)} - \$27,000 = \$53,000$.
- "Total Assets (Excluding Cash)": $\$20,833 + \$150,000 + \$30,000 + \$53,000 = \$353,833$.

"Liabilities & Equity":

- "Accounts Payable": Item 7 says "40 days' purchases". First we find Purchases: $\text{COGS } \$600k + \text{Ending Inv } (150k) - \text{Beginning Inv } (150k) = \$600,000$.
Calculation: $\$600,000 / 360 \times 40 = \$66,667$. (Rounded).
- "Income Tax Payable": Item 9 says "100% of year's taxes" (the tax expense we calculated above).
Calculation: $\$10,500 \times 1.00 = \$10,500$.
- "Common Stock": Item 10 value is "\$100,000".
- "Retained Earnings (Ending)": Beg Balance (\$75,000) + Net Income (\$31,500) - Dividends (\$7,875).
Calculation: $\$75,000 + \$31,500 - \$7,875 = \$98,625$.

"The Plug (Bank Loan or Cash):"

← Under Current liabilities "Working Capital loan"

The balance sheet must balance ($\text{Assets} = \text{Liabilities} + \text{Equity}$). We solve for the missing "Plug" figure in Assets.

- "Total Liabilities & Equity": $\$66,667 \text{ (AP)} + \$10,500 \text{ (Tax Pay)} + \$100,000 \text{ (Stock)} + \$98,625 \text{ (RE)} = \$275,792$.
- "Calculation": Total L&E: $\$275,792$ - Assets without Cash: $\$353,833 = \$78,041$.

"Result for Scenario 1:" You have a positive surplus of \$78,041, which is recorded as "Cash (Bank Loan)". The company has extra cash on hand.

Scenario 2: Sales Manager's Estimates (\$1,000,000 Sales)

The sales manager wants to hit \$1M in sales. Let's see if that actually helps the company or just causes a deficit.

"Income Statement Changes":

- "Sales": "\$1,000,000".
- "COGS (80%)": "\$800,000".
- "Gross Profit": "\$200,000".
- "Operating Expenses":
 - "Salary": \$42,000 (Fixed).
 - "Depreciation": \$6,000 (Fixed).
 - "Other OpEx (8% of Sales)": $\$1,000,000 \times 0.08 = \$80,000$.
 - "Total OpEx": $\$42k + \$6k + \$80k = \$128,000$.
- "Operating Income": $\$200,000 - \$128,000 = \$72,000$.
- "Tax (25%)": $\$72,000 \times 0.25 = \$18,000$.
- "Net Income After Tax": $\$72,000 - \$18,000 = \$54,000$.
- "Dividends (25% of NI)": $\$54,000 \times 0.25 = \$13,500$.
- "Retained Earnings (Ending)": Beg (\$75,000) + NI (\$54,000) - Div (\$13,500) = "\$115,500".

"Balance Sheet Changes":

- "Accounts Receivable": 10 days of \$1M sales.
Calculation: $\$1,000,000 / 360 \times 10 = \$27,778$.
- "Inventory": 90 days (1/4 year) of new COGS (\$800k).
Calculation: $\$800,000 / 4 = \$200,000$.
- "Land & Net Buildings": Unchanged at "\$93,000" total (\$30k + \$63k).
- "Total Assets (Excl Cash)": $\$27,778 + \$200,000 + \$93,000 = \$320,778$.

261,473 ?

"The Plug Calculation":

- "Accounts Payable": We need Purchases first. $\text{COGS } \$800k + \text{End Inv } (200k) - \text{Beg Inv } (150k) = \$850,000$.
AP Calculation: $\$850,000 / 360 \times 40 = \$94,444$. (Rounded).
- "Income Tax Payable": $\$18,000 \text{ (Expense)} \times 1.00 = \$18,000$.
- "Total Liabilities & Equity (Excl Cash)": $\$94,444 + \$18,000 + \$100,000 \text{ (Stock)} + \$115,500 \text{ (RE)} = \$327,944$.
- "Plug": Total Assets (\$320,778) - Total L&E (\$327,944) = $-\$7,166$.

"Result for Scenario 2:" The surplus cash is even ~~problem~~ $-\$7,166$, despite the higher sales volume, because you grew your inventory faster than you paid off your debts to suppliers.

+ 3907

taking out more
money to support
more profit

straight up
increment

Always State Assumptions

Assume Cash 0 in new balance sheet

negative plug => we have excess money

